SPE-229035-MS

Optimizing Horizontal Wells Trajectory to Improve Artificial Lift Reliability and Reduce Deferment

A. Al Lawati, B. A. Salem, M. Al Harthi, S. Al Kindi, A. Al Zadjali, A. Al Amri, Z. Al Rawahi, O. Al Hamdani, W. Al Yaaqoubi, and R. Al Kindi, Petroleum Development Oman, Muscat, Oman

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

Reliability enhancement of the oil producing wells with artificial lift and reducing deferment is an extremely critical element in the oil field industry where drilling wells with the proper design and configuration can further improve their performance and production gains. In the new oil operations, having the perfect design for each well while meeting all production objectives and reservoir targets provides excellent production rates with minimum costs to sustain production. In horizontal oil producers that require artificial lifting systems, drilling plan and operations must always consider pump requirements in order to provide an optimized and reliable well that will have continuous production with the lowest possible maintenance costs.

Every artificial lifting system has its own requirements and criteria to operate properly depending on the components and configurations it contains. In the South of Oman, where low to heavy oil is being produced, the selection of the proper artificial lifting system is extremely critical along with providing a well that can utilize the artificial lifting system properly. Artificial lift system failure statistics show that majority of the Rod, Tubing, and pump failures occur due to trajectory concerns in the deviated section (KOP to tangent) and the tangent section. Additionally, new oil wells that are being drilled are reported with 40% of wells with trajectory concerns where 50% of them with early failures.

The presented paper focuses on providing a Fail-Less strategy for the PCP system by focusing on the importance of trajectory planning and execution in the new oil horizontal PCP wells to reduce the number of wells with trajectory concerns from 40% to less than 5% along with reducing the failure rates of the new wells from 50% to 0%. Additionally, it explains the value of providing reliable wells along with the savings in cost and reduction in possible deferment. An optimized trajectory drilling plan and drilling procedure can be utilized to produce reliable PCP oil producers that can sustain excellent production with lowest possible maintenance cost and deferment.

Introduction

Progressive cavity pumps (PCPs) are widely used in new oil horizontal wells in the South of Oman since PCPs meet the artificial lift selection criteria in the South Clusters. More than three thousand PCP wells are currently operating in the South Clusters with an average of 200 new PCP wells drilled every year where majority of the wells are horizontal oil producers. These wells target shallow sandstone reservoirs that produce high water cut (+90%) and high solid contents (+200 g/M3) with oil varying from light to heavy. Reservoir completion selection varies also where some wells are completed with open hole completion, some are completed with perforated cased hole completion, and some are completed with sand control.

The PCP System consists of Drive Hear, Motor, Tubing String, Rod String, and the Progressive Cavity Pump that contains the Rotor and the Stator as shown in Fig.1. The system operates by providing the required energy to the Motor which rotates the Drive Head that holds the Rod String. The Rod String could be designed using Sucker Rods that require couplings between each Rod joint, or Continuous Rods that do not use any couplings and run from the top Polished Rod up to the Rotor. The Rods rotate resulting in rotating the Rotor inside the Stator of pump causing cavities that displace the fluid from the wellbore to the inside of the Tubing and eventually to the surface. For continuous production from the well, the Rod String must rotate continuously inside the Tubing String where both could be damaged if there was high contact rate between the Rod and the Tubing. Additionally, higher rates of damage occur on the Tubing with high water cut and sand content resulting in Rod or Tubing failures since there is less lubrication and higher rates of erosion. Considering the deign and operation requirements of the PCP, the well trajectory plays a significant role on the reliability of the well and its capability on sustaining long run life without required maintenance.

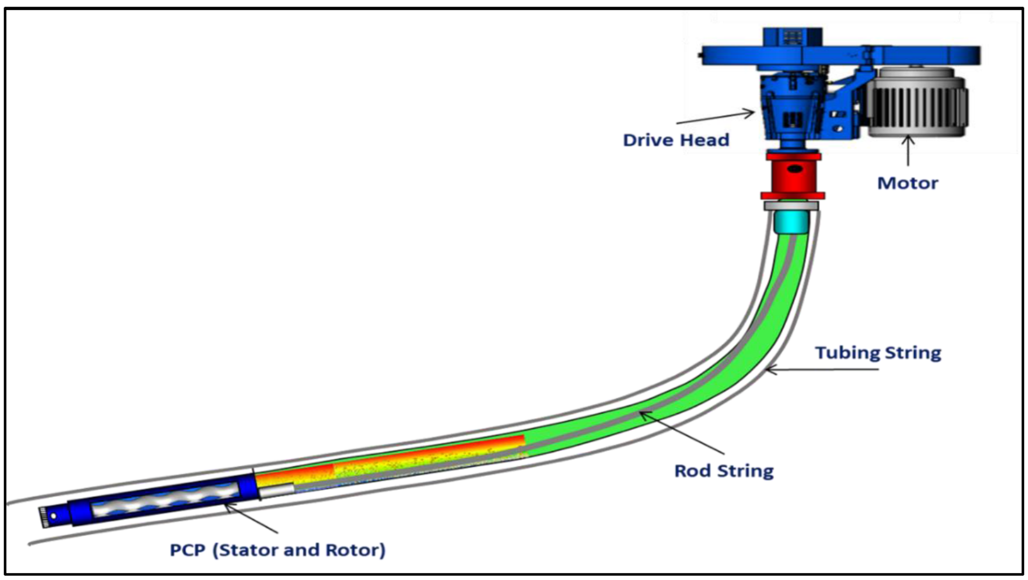

Figure 1—PCP System and Components

Problem Statement and Analysis

Multiple studies were conducted on the collected failure statistics and analysis to determine the reason of failed PCP systems. From the failure's information, it was found that majority of the failures are related to Rod or Tubing failures. Rod failures reported are related to parted Rod String in the Rod body or unscrewed Rod in the coupling connections in the PCP system as shown in Fig.2.a. Sucker Rods and Continuous Rods have both been used for the PCP wells but still reported with failures in Rod body or couplings.

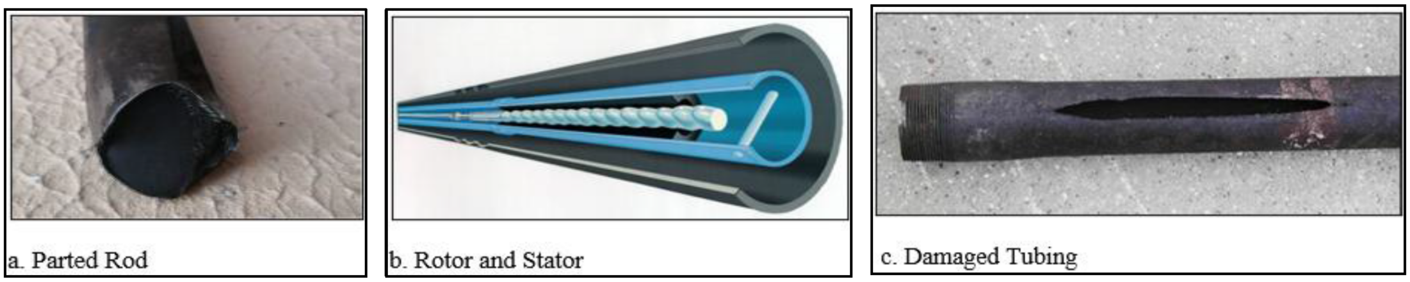

Figure 2—Parted Rod, Rotor and Stator, Damaged Tubing

Tubing failures are related to parted or damaged Tubing resulting to a hole in Tubing as shown in Fig.2.c. Failure findings confirmed that such failures occur due to continuous contact between the Rod String and the inside section of the Tubing String during the continuous Rod rotation inside the Tubing String. For the wells that use Sucker Rods in the design, the Tubing failures are reported where the couplings are usually placed since it is the highest contact location. As for Continuous Rods, the Tubing failures are usually the area with the highest deviation in the well where Rod/Tubing contact is the highest.

After collecting the information about the wells that suffer with this type of failures in less than one year of operating, it was found that wells drilled with long deviated section (KOP to tangent) fail faster compared to wells with shorter than 200 m deviated section. Early failures occur due to higher area with Rod/Tubing contact, Rod String tends to put more stress on one part of the Tubing resulting in failure, and failure could occur in Rod (parted Rod), or Tubing (hole in Tubing)

Additionally, it was found that wells which have high Dogleg Severity (DLS higher than 1 Deg/30m) in the tangent section, where the pump is installed, fail faster compared to wells that have lower DLS in the tangent section. Also, wells that have a tangent section that is lower than 40 m in length tend to fail faster as the length of the tangent section should consider the length of the pump and the length of the orbital Tubing (Tubing String that is installed above PCP Stator). Since the PCP requires continuous rotation of the Rotor inside the Stator as shown in Fig.2.b, any inclination or change in azimuth in the tangent section results in having the pump not completely straight where misalignment of the Rotor/Stator causes pump failure. Failure information shows that failures could occur in PCP (parted Rotor, damaged elastomer) due to un-even rotation. Furthermore, failures could occur in the orbital Tubing where continuous Rod rotation and eccentricity movement of the Rod while rotating would damage the Tubing String resulting in hole in Tubing failure.

A study was conducted on two areas that were severely affected by the PCP failures, and both were showing high rates of failure index reaching up to 42.7% in one of the areas with a mean time between failure (MTBF) reaching as low as 494 days. Additionally, a study conducted on the new oil PCP wells drilled and had a failure within less than a year which showed that 84% of the wells had a failure in that area (21 failures in 25 wells drilled). Results of both studies are shown in Fig.3 for area 1 and Fig.4 for area 2. Each failure comes with an estimated cost of recovery reaching $100k. Fig.5 shows the PCP failure index in the South Clusters indicating high failure rates for PCP wells.

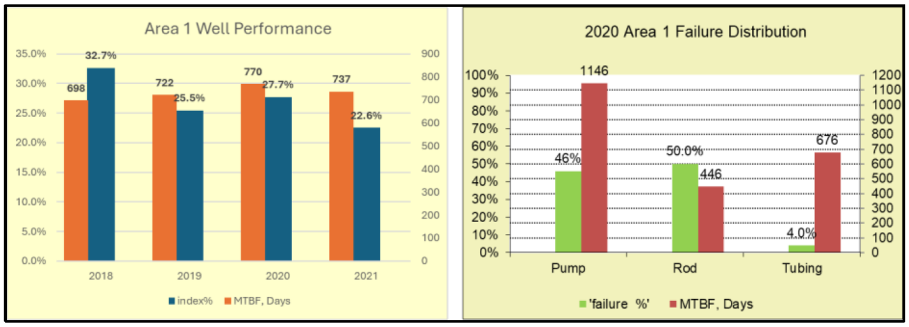

Figure 3—Area 1 PCP Failure Information

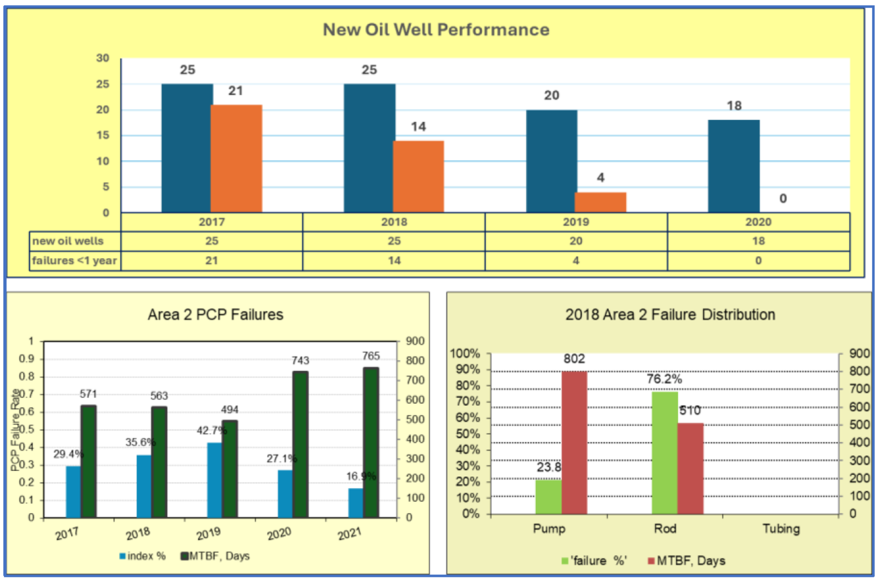

Figure 4—Area 2 PCP Failure Information

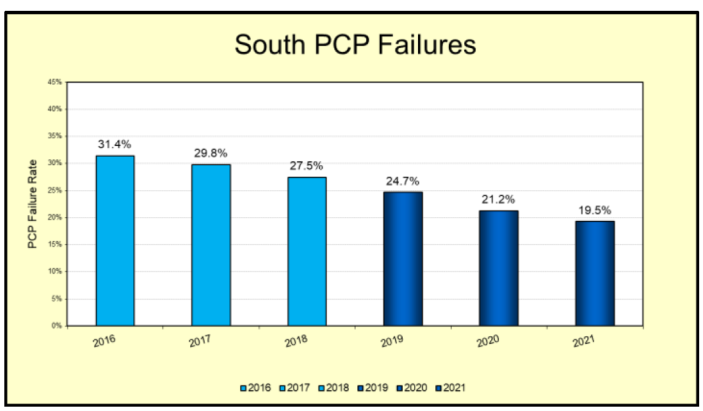

Figure 5—PCP Failure Index in the South Clusters

Failure statistics performed on the new oil wells that were drilled found that 40% of the wells drilled in 2020 had trajectory concerns where majority of the wells had faced at least one of the following issues:

- Deviated section (KOP to tangent) was longer than 200 m of length.
- Tangent section DLS is higher than 1 Deg/30m.
- Tangent section length was lower than 40 m.

As a result of such trajectory concerns, 50% of the wells suffered with early pump failures (failure in less than 1 year). The remaining wells eventually faced similar failures and had to be operated with extra caution to avoid failures. All the wells that had such trajectory concerns had optimization limitations where maximum allowable speed to operate the PCPs were limited, and PID function was not accepted as an optimization method (PID function optimizes the PCP running speed based on the available intake pressure and operating conditions).

As a solution for the wells that had trajectory concerns, new technologies and upgraded materials were used such as the following:

- Use of upgraded Rod materials:
  - Alpha Rods (300% increased cost).
  - Rod Centralizers and Molded Guides (150% increased cost).
- Use of upgraded Tubing materials:
  - GRE Tubing (150% increased cost).
  - Polyline Tubing (150% increased cost).

Upgrading the materials displayed excellent improvement in the PCP system performance where more than four hundred wells were completed using GRE Tubing and Molded Guide Sucker Rods with 99% success rate and increased PCP system run life. However, upgrading the materials comes with an increased cost of materials and intervention.

Considering that two hundred wells are drilled yearly, with the history of 40% of wells with trajectory concerns, and 50% of them with early failures, 40 wells are expected to have early pump failures yearly with an estimated required cost of $5M every year to restore them back to production. Also, the wells might require multiple interventions where the cost of intervention would accumulate resulting in higher costs on the newly drilled wells to maintain the production. By estimation, the wells that were drilled with trajectory concerns would need an average of five interventions within a 10-year span. This would result in an estimated cost of $50M to maintain the production from the wells that are drilled each year.

Well with trajectory concerns have higher failure rates, strict optimization limitations, and higher cost of intervention to restore due to higher number of required interventions and cost of materials. Additionally, higher rates of oil deferment are recorded for the wells since they are new wells, have system failures within less than a year for multiple wells, and require time to restore back to production.

Fail-Less Strategy

To further improve the performance of the PCP systems in the South of Oman, a Fail-Less strategy was initiated by the Artificial Lifting Technical Team. The strategy covers all aspects of the PCP system to further improve the system performance, increase PCP run life, decrease number of failures, reduce oil deferment, improve failure index, and improve PCP system reliability. Part of the Fail-Less strategy focused on the wells that were drilled already with trajectory concerns to further improve their performance and reliability using new technologies and improved materials. The results from the improvements using the GRE and Polyline Tubing with Molded Guide Sucker Rods and Alpha Rods showed excellent results.

Another part of the Fail-Less strategy focused on the new oil wells that are planned to be drilled within the next years to produce reliable and optimized wells that can sustain production without pump failures or upgraded materials. Using the findings from the failure analysis and statistics while aligning with the objectives from the Fail-Less strategy, the following criteria was established for the trajectory requirements while planning and drilling the new oil PCP horizontal wells:

- Requirements for the build-up section before tangent section for PCP wells:
  - Maximum of 200 m from KOP to tangent section.
  - DLS less than 6.5 Deg/30m.
- Requirements for the tangent section in PCP wells:
  - Minimum of 40 m tangent section length (60 m is preferred).
  - As deep as possible closer to the targeted reservoir.
  - Max DLS of 1 Deg/30m.

Failure statistics confirms that wells with shorter than 200 m deviated section (KOP to tangent) have better system run life. Additionally, obtaining a straight tangent section would ensure proper alignment of PCP inside the tangent section for proper operations with lower risk to pump failure.

PCP modeling was performed also to validate the improvement in the trajectory criteria, and it confirmed that increasing the DLS by 1 Deg/30m would exponentially increase the Rod/Tubing contact and reduce the expected system run life. C-FER Application used for PCP system designing showed the below results in Fig.6 which indicate a sever increase in Rod/Tubing contact with very minor increase in DLS.

DLS	1227.6 mKB	1236.7 mKB
1	230.6	218.6
2	834.7	935.7
3	1364	1365

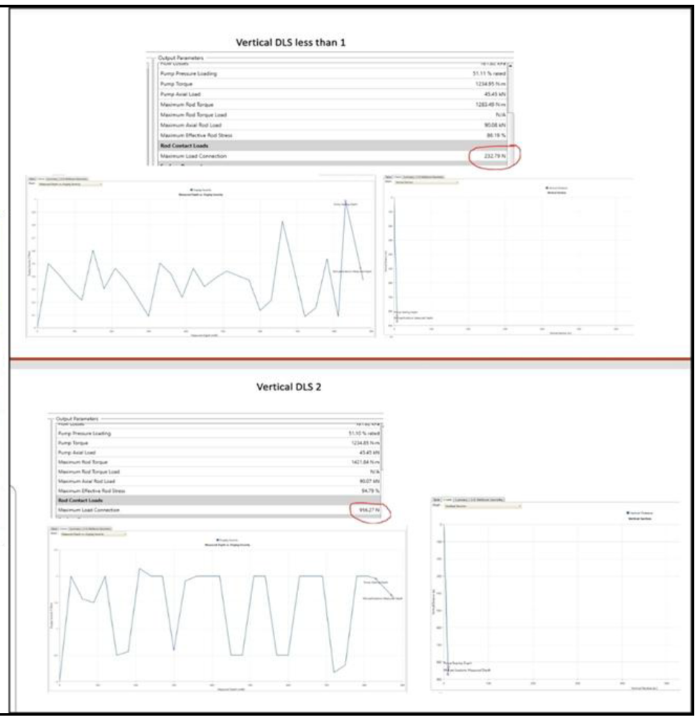

Figure 6—PCP Software Analysis Between Different Deg/30m of DLS and Rod/Tubing Contact Load

During the software analysis, the following results were collected for one of the wells in the study:

- DLS at 1 Deg/30m results 233 N Rod/Tubing contact load.
- DLS at 2 Deg/30m results 956 N Rod/Tubing contact load.
- Rod/Tubing contact load at 200 N results in 80% Tubing wall loss in the first year.
- Rod/Tubing contact load at 400 N results in 100% Tubing wall loss in the less than a year (8-10 months).

- Rod design could reduce the Rod/Tubing contact load if speed is controlled, or Centralizers were utilized.

Methodology

In order to implement the Fail-Less strategy on the new oil PCP horizontal wells, the strategy requirements must be implemented to optimize well planning and design, drilling procedure, drilling cost, and drilling time to accommodate for the additional cost and time required. In well planning and design, the following requirements must be considered to achieve an optimized design in the build-up section before tangent section and the tangent section for PCP wells:

- Shallow Kick-Off Point (KOP) is recommended to reach the targeted reservoir (shallow reservoirs).
- Wells designed with PCP tangent section after the KOP by less than 200 m.
- Deviated section (KOP to tangent) planned with less than 6.5 Deg/30m DLS.
- Tangent section located within 100 m TVD from the reservoir.
- Tangent section designed with a minimum of 40 m length (60 is recommended).
- Tangent section designed with 0 Deg/30m DLS.
- Trajectory plan review with Artificial Lift Technical Team to confirm requirements.

During drilling operations for the new oil wells, the following steps must be followed to confirm optimized trajectory is obtained for well. Optimized drilling operations flow chart is shown in Fig.7.

- Well trajectory plan must be confirmed to have less than 200 m of deviated section (KOP to tangent) before drilling operations start.
- KOP depth must be confirmed to meet planned trajectory.
- DLS for the deviated section must be maintained below 6.5 Deg/30m.
- Tangent section should be planned with a minimum of 40 m and 0 Deg/30m DLS.
- Tangent section must be drilled with controlled ROP (Rate of Penetration) with a minimum of 5 m/hr and a maximum of 10 m/hr.
- Tangent section trajectory must be checked to confirm that its meeting requirements (minimum of 40 m and less than 1 Deg/30m DLS).
- If trajectory is not meeting requirements, reaming, wiping, and circulation must be done to reduce the DLS below 1 Deg/30m.
- Once trajectory is confirmed to meet requirements, drilling the remaining of the well can commence and Casing can be RIH.
- To confirm the results of the trajectory, Gyro survey can be run for the well.

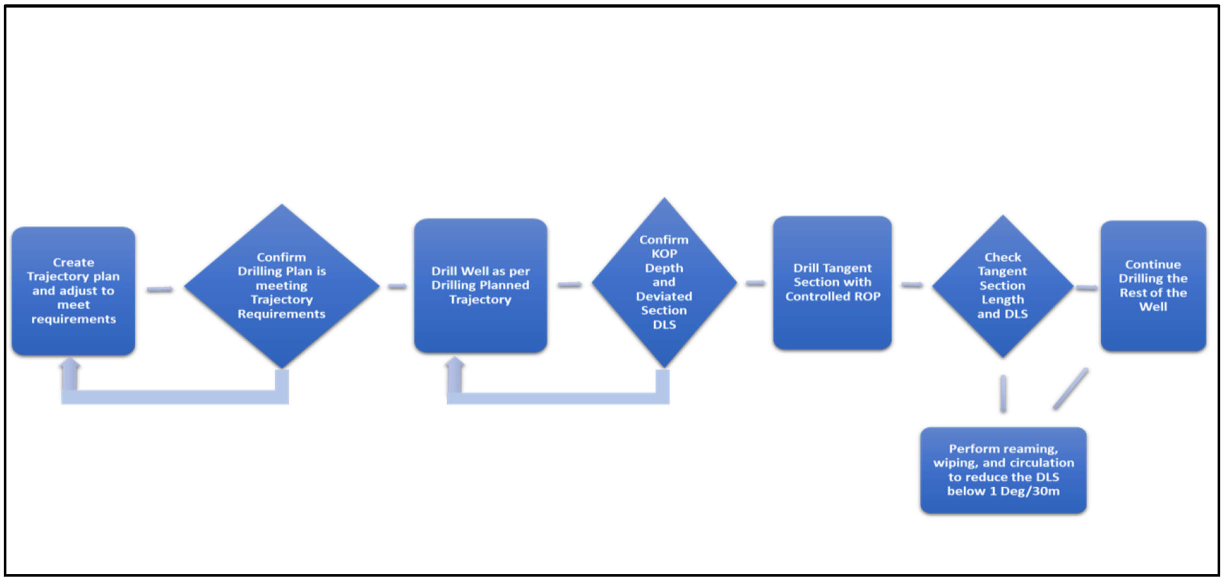

Figure 7—Optimized Drilling Operations Flow Chart

By adding these additional steps in the procedure to drill the well up to the end of the tangent section, additional time and cost are required. Drilling with considerations not to exceed 6.5 Deg/30m DLS in the deviated section and a controlled ROP for a minimum of 40 m in the tangent section requires an additional estimated 18 hours of operations which is directly connected to increasing drilling costs and time. With the average of two hundred wells drilled yearly, an estimated additional cost of $6.6M would be required to meet trajectory requirements for the PCP wells along with the additional 3600 hours.

Another challenge for the implementation of the optimized trajectory was achieving 1 Deg/30m DLS for the tangent section as it was not guaranteed for the wells due to the complexity of the areas drilled and the sensitivity of the DLS calculation which changes with any slight change in Inclination/Azimuth while drilling. Achieving less than 200 m of deviated section (KOP to tangent) is possible with DLS lower than 6.5 Deg/30m since the section is longer the limitations are wider.

To overcome the challenges to achieve the optimized trajectory for the new oil wells, the following steps were utilized:

- Secure required budget for the additional time and cost required for drilling operations.
- Continuous drilling operation follow-up by Production Technologist, Production Geologist, and Artificial Lift Technical Team.
- Continuous support from Management Team and Artificial Lift Technical Team.
- Excellent performance from Well Engineering and Rig Crew.

Results and Comparison

As a result of implementing the Fail-Less strategy by using the optimized trajectory requirements for the new oil PCP wells in the South of Oman, the number of early failures related to Tubing or Rod reduced from 50% to 0% recorded since the implementation. Also, the number of wells drilled with trajectory concerns reduced from 40% to 4% resulting in wells with excellent performance and no requirements for further interventions to maintain them. The trajectory optimization resulted in reliable wells that have better run life and improved failure index since no early failures or deferment are occurring. By estimating that the wells would require two interventions within a 10-year span, the total estimated cost to maintain the wells reduced from $50M to $20M or less, especially if the wells are operated within the Operating Envelope (minimum and maximum allowable running speed, Rod torque, and intake pressure that PCPs should operate within). Such reduction in workover interventions results in lower number of failures in the PCP wells, lower operating costs required to operate the wells (cost of workover Hoist, materials, and time) and lower rate of oil deferment due to PCP failures. Fig.8 is showing the results of implementing the optimized trajectory for the horizontal PCP wells.

Fail-Less Strategy Implementation	Before Implementation	After Implementation
Wells with Trajectory Concerns	40%	4%
Rate of Early Failures	50%	0%
Number of Interventions required per Well (10 Years)	5 Interventions	2 Interventions
Maintenance Intervention Costs in 10 Years	$50M	$20M

Figure 8—Comparison of Implementing the Optimized Trajectory for the Horizontal PCP Wells

In comparison, the total cost to implement the optimized horizontal wells trajectory is estimated at $6.6M for two hundred wells. However, it would save around $30M cost of intervention in the future with wells drilled under the Fail-Less strategy. Furthermore, lower number of wells would require upgrading the materials used in the completion (Alpha Rods, GRE Tubing, Polyline Tubing, and Molded Guide Sucker Rods) since the trajectory is optimized.

The implementation of the Fail-Less strategy started in 2018 where the results started showing excellent improvements during the next years as shown in Fig.5 where the failure index started reducing during the years after implementation from 31.4% down to 19.5%. area 1 took longer time to obtain the improved results, since wells from previous years were still facing issue, but the failure index did improve from 32.7% down to 22.6% in three years as shown in Fig.3. As for area 2, the improvement observed was faster where the number of early failures in new oil wells dropped from 84% to 0% in four years and the failure index went down from 42.7% to 16.9% after the implementation of the Fail-Less strategy as shown in Fig.4.

With the reduction of the number of require interventions to restore the wells due to implementing the optimized trajectory requirements for new oil PCP wells, the expected deferment from these reliable wells would reduce. Wells are not going to be closed frequently due to any possible failures (Pump, Tubing, or Rod) and will produce continuously resulting in lower deferment and higher production rates for the wells. Moreover, wells can be optimized easily (when drilled with optimized trajectory) with no limitations of PID utilization and maximum allowable running speed.

Conclusion and Recommendations

In conclusion, optimizing the trajectory and drilling new oil PCP wells under the Fail-Less strategy provides a reliable PCP system that has exceptionally low possibility of failure in the Rod or Tubing. Such system is capable of operating without any required interventions to restore the production resulting in higher rates of production, lower rates of deferment, and lower cost of maintenance. The implementation of the optimized trajectory requirements is essential to the new drilled wells to produce an improved sustainable system with sustainable production and efficient PCP optimization.

With the excellent results observed in the implementation of PCP wells trajectory optimization, the same concept was implemented in other artificial lift systems in the South such as Beam Pumps (BP) and Electrical Submersible Pumps (ESP) where the criteria would be different depending on the mechanism of the system.

Acknowledgment

The authors of this project extend their sincere gratitude to the management teams at Petroleum Development Oman for their invaluable support for this project and the implementation while acknowledging the excellent outcome from its results.

References

Al Ajmi, A., Kulkarni, R., Al Farsi, T., Al Busaidi, I., Borkent, E., Al Kaabi, I., Al Jahdhami, A., Al Shukaili, G., AL Ismaili, A., Al Kindi, S., Al Noumani, H., and N. Al Azri. "Enhancing WRFM Operational Excellence by Addressing High Artificial Lift Failure Index Through Comprehensive Fail-Less Strategy - A Case Study in a High Viscous Onshore Oil Field." Paper presented at the ADIPEC, Abu Dhabi, UAE, SPE-222191-MS, November 2024. (https://doi.org/10.2118/222191-MS)

Al Asmi, Azzan, Landgraf, Christian, Al Abri, Hilal, Montilla, Benigno, Caridad, Jose, Perez, Pedro, and William Wyninegar. "Improving Run Life in PCP Wells with CO2 and H2S Concentrations Using Tungsten-Alloy-Coated Rods and Couplings." Paper presented at the SPE Middle East Artificial Lift Conference and Exhibition, Manama, Bahrain, SPE-206928-MS, October 2022. (https://doi.org/10.2118/206928-MS)

Al-Sidairi, Nada, Osman, Mahmoud, Oritola, Nasir, Ahmed, Farsi, Zuhaimi, Khalid, Amri, Mohsin, Hinai, Ghalib, and Majid Al-Mahrooqi. "Overcoming Artificial Lift Failures in Polymer Flooding Oil Field in the South of Oman." Paper presented at the SPE EOR Conference at Oil and Gas West Asia, Muscat, Oman, SPE-190390-MS, March 2018. (https://doi.org/10.2118/190390-MS)

Clarke, Fred "Progressing Cavity Pump Application in 40 API Light Oil-The Cooper Basin Field Study." Paper presented at the SPE Progressing Cavity Pumps Conference, Calgary, Alberta, Canada, SPE-165664-MS, August 2013. (https://doi.org/10.2118/165664-MS)

Denney, Dennis. "Overcoming Challenges of Drilling High-Dogleg-Severity Curves." *J Pet Technol* **63** (2011): 79–83, SPE-1111-0079-JPT. (https://doi.org/10.2118/1111-0079-JPT)

Khadav, Sandeep, Agarwal, Shubham, Kumar, Piyush, Pandey, Nimish, Parasher, Arunabh, Kumar, Sanjeev, Agarwal, Vinay, and Shobhit Tiwari. "System Run Life Improvement for Rod Driven PCP in High Deviation Well." Paper presented at the *SPE Artificial Lift Conference and Exhibition - Americas*, The Woodlands, Texas, USA, SPE-190948-MS, August 2018. (https://doi.org/10.2118/190948-MS)

Matthews, C. M., and L. J. Dunn. "Drilling and Production Practices To Mitigate Sucker Rod/Tubing Wear-Related Failures in Directional Wells." *SPE Prod & Fac* **8**, SPE-22852-PA, (1993): 251–259. (https://doi.org/10.2118/22852-PA)

Toelsie, S., Mohan, A., Kisoensingh, S., and P. Goerdajal. "Overcoming Nearly Three Decades of PCP Experience in Shallow Heavy Oil Field." Paper presented at the SPE Progressing Cavity Pumps Conference, Calgary, Alberta, Canada, SPE-165067-MS, August 2013. (https://doi.org/10.2118/165067-MS)

Wiltse, D. J., and B. A. Weir. "Eliminating Sucker Rod Couplings Can Reduce Progressing Cavity Pump Operating Costs." Paper presented at the SPE International Heavy Oil Symposium, Calgary, Alberta, Canada, SPE-30275-MS, June 1995. (https://doi.org/10.2118/30275-MS)